

sm-admin SFTP Setup Guide

Create SFTP user with access only to the HTML directory on an EC2 instance

User to create	sm-admin
Group to create	sftpusers
Allowed folder	/html inside SFTP, mapped to /var/www/html on the server

Beginner-friendly command guide with explanations

1. What We Are Creating

Final result: User sm-admin will connect using SFTP. After login, the user will see only /html. That /html folder will actually point to /var/www/html on the EC2 server.

Visual flow:

```

SFTP client
  |
  v
sm-admin user
  |
  v
Chroot jail: /sftp/sm-admin
  |
  v
Visible folder: /html
  |
  v
Actual website folder: /var/www/html

```

2. Install And Start SFTP

SFTP usually comes with OpenSSH server. You install SSH server, and SFTP becomes available through it.

sudo apt update	
Use	Refreshes Ubuntu/Debian package information.
Why	We use it before installing packages so the server knows the latest available package versions.

sudo apt install -y openssh-server	
Use	Installs OpenSSH server.
Why	SFTP support is built into OpenSSH, so this gives the server SSH and SFTP capability. The -y automatically answers yes during installation.

sudo systemctl enable --now ssh	
Use	Starts SSH now and enables it after reboot.
Why	On Ubuntu/Debian the service name is usually ssh. If sshd.service gives alias error,

<code>sudo systemctl enable --now ssh</code>	
Use	Starts SSH now and enables it after reboot.
	use <code>ssh.service</code> like this.

<code>sudo systemctl status ssh</code>	
Use	Shows whether SSH service is running.
Why	If SSH is not running, SFTP connections will not work.

<code>sudo sshd -T grep -i sftp</code>	
Use	Checks OpenSSH effective config and searches for SFTP subsystem.
Why	If you see <code>subsystem sftp</code> or <code>internal-sftp</code> support, SFTP is available.

3. Create Group And User

<code>sudo groupadd -f sftputers</code>	
Use	Creates the Linux group named <code>sftputers</code> .
Why	We use one group to control which users can edit the HTML directory. The <code>-f</code> avoids an error if the group already exists.

<code>sudo useradd -g sftputers -d /html -s /usr/sbin/nologin sm-admin</code>	
Use	Creates user <code>sm-admin</code> with primary group <code>sftputers</code> , SFTP start folder <code>/html</code> , and no normal shell login.
Why	We want <code>sm-admin</code> to use SFTP only, not normal SSH terminal access. The <code>/html</code> home path is what the user sees after SFTP login.

<code>sudo passwd sm-admin</code>	
Use	Sets the password for <code>sm-admin</code> .
Why	The user needs a password if you are allowing password-based SFTP login.

4. Give Group Access To HTML Directory

<code>sudo chgrp -R sftpusers /var/www/html</code>	
Use	Changes group ownership of /var/www/html and everything inside it to sftpusers.
Why	Because sm-admin belongs to sftpusers, group ownership lets us grant access without making sm-admin the owner of the whole website folder.

<code>sudo chmod -R g+rwX /var/www/html</code>	
Use	Gives the group read and write access. Capital X gives directory-enter permission only where needed.
Why	This lets sm-admin upload, edit, and list files inside the HTML directory.

<code>sudo find /var/www/html -type d -exec chmod g+s {} \;</code>	
Use	Finds all directories under /var/www/html and applies setgid to them.
Why	New files/folders created later will keep group sftpusers, so future uploads do not break permissions.

5. Create SFTP Jail And Map HTML Directory

<code>sudo mkdir -p /sftp/sm-admin/html</code>	
Use	Creates the SFTP jail folder and the html folder inside it.
Why	The -p creates missing parent folders too. This folder structure is where sm-admin will be locked.

<code>sudo chown root:root /sftp /sftp/sm-admin</code>	
Use	Makes root the owner of the chroot jail folders.
Why	OpenSSH requires the chroot directory and its parents to be owned by root and not

<code>sudo chown root:root /sftp /sftp/sm-admin</code>	
Use	Makes root the owner of the chroot jail folders.
	writable by the SFTP user.

<code>sudo chmod 755 /sftp /sftp/sm-admin</code>	
Use	Sets permissions so root can write, and others can read/enter.
Why	The chroot folder must not be writable by sm-admin, otherwise OpenSSH rejects login for security reasons.

<code>sudo mount --bind /var/www/html /sftp/sm-admin/html</code>	
Use	Mounts the real website folder into the SFTP jail.
Why	This is the key mapping. sm-admin sees /html, but the actual files remain in /var/www/html.

6. Add Permanent Mount In /etc/fstab

This is not a terminal command. Open the file first:

```
sudo nano /etc/fstab
```

Then paste this plain-text line at the bottom:

```
/var/www/html /sftp/sm-admin/html none bind 0 0
```

Why: mount --bind is temporary unless it is added to /etc/fstab. This line makes the mapping come back automatically after reboot.

Save in nano: Ctrl + O, press Enter, then Ctrl + X.

7. Add SSH SFTP Rules

This is also not a terminal command block. First open the main SSH config file:

```
sudo nano /etc/ssh/sshd_config
```

Paste this plain-text config at the bottom of the file:

<pre>Match User sm-admin ChrootDirectory /sftp/sm-admin ForceCommand internal-sftp -d /html PasswordAuthentication yes AllowTcpForwarding no X11Forwarding no PermitTTY no Match all</pre>	
Line	Meaning
Match User sm-admin	Applies these SSH rules only to sm-admin.
ChrootDirectory /sftp/sm-admin	Locks the user inside /sftp/sm-admin so the user cannot browse the whole server.
ForceCommand internal-sftp -d /html	Forces SFTP mode and starts the user inside /html.
PasswordAuthentication yes	Allows password login for this user.
AllowTcpForwarding no	Blocks SSH tunneling features that are not needed for SFTP.
X11Forwarding no	Blocks graphical forwarding, not needed for SFTP.
PermitTTY no	Blocks terminal access.
Match all	Ends the Match block so the rules do not leak into other users.

Save in nano: Ctrl + O, press Enter, then Ctrl + X.

8. Test SSH Config And Reload

sudo sshd -t	
Use	Tests SSH configuration syntax.
Why	Always run this before reload. If the SSH config has a mistake, this command shows it before you risk breaking SSH access.

<code>sudo systemctl reload ssh</code>	
Use	Reloads the SSH service.
Why	This applies the new SFTP rules without fully restarting the service.

9. Test SFTP Login

<code>sftp sm-admin@<EC2_PUBLIC_IP></code>	
Use	Connects to the EC2 server as sm-admin using SFTP.
Why	Replace <EC2_PUBLIC_IP> with your EC2 public IP. After login, the user should see only /html.

10. Full Command Checklist

Run these in order on the EC2 terminal:

```

sudo apt update
sudo apt install -y openssh-server
sudo systemctl enable --now ssh
sudo systemctl status ssh
sudo sshd -T | grep -i sftp

sudo groupadd -f sftpusers
sudo useradd -g sftpusers -d /html -s /usr/sbin/nologin sm-admin
sudo passwd sm-admin

sudo chgrp -R sftpusers /var/www/html
sudo chmod -R g+rwX /var/www/html
sudo find /var/www/html -type d -exec chmod g+s {} \;

sudo mkdir -p /sftp/sm-admin/html
sudo chown root:root /sftp /sftp/sm-admin
sudo chmod 755 /sftp /sftp/sm-admin
sudo mount --bind /var/www/html /sftp/sm-admin/html

sudo nano /etc/fstab
sudo nano /etc/ssh/sshd_config

sudo sshd -t
sudo systemctl reload ssh

```

11. Important Notes

- Commands shown in code blocks with `sudo` are run in the EC2 terminal.
- Plain-text config lines are pasted inside files opened with `nano`.
- EC2 Security Group must allow inbound TCP port 22 from your IP, otherwise SFTP will not connect.
- If your real HTML folder is not `/var/www/html`, replace `/var/www/html` everywhere with the correct path.
- Password login is simple for testing, but SSH key login is safer for production.